

National University of Computer and Emerging Sciences, Lahore Campus



Course: Electro Mechanical Systems
Program: BS(Electrical Engineering)
Duration: 60 Minutes
Paper Date: 04-Nov-17
Section: ALL
Exam: Mid2 Examination

Course Code: EE221
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Total Marks: 60
Weight: 15%
Page(s): 6
Questions: 3

Name: [REDACTED] Roll No.: [REDACTED] Section: [REDACTED]

Instruction/Notes:

- > Don't ask any questions; in case of an ambiguity, use valid assumptions if needed.
- > State the units of each value where necessary.

Q1 A ^{VT} 2500-V, ^S 1200-kVA, 0.8-PF-lagging, 60-Hz, 4 pole, Y-connected synchronous generator, has a synchronous reactance of 1.2Ω and an armature resistance of 0.2Ω . At 60 Hz, its friction and windage losses are 30 kW, and its core losses are 20 kW. The field circuit has a dc voltage of 220 V, and the maximum I_F is 10 A. The resistance of the field circuit is adjustable over the range from 22 to 200 Ω . The field current has been adjusted to 5.5A and a delta connected load having per phase impedance of $18.186 + j10.5$ ohm is connected to the generator. OCC of the generator is given below. [20] CLO 04

Find out the following quantities.

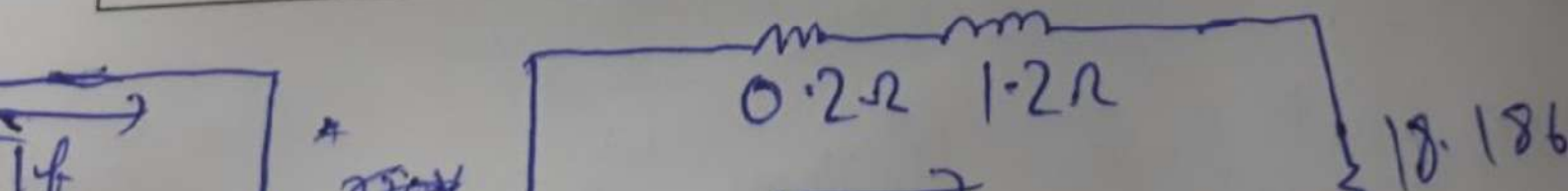
- 1) voltage regulation
- 2) efficiency
- 3) output torque of prime mover

V_R $|E_{ind}| - V$
 V_ϕ

I_F (A)	2	4	6	8	10
$V_{T,NL}$ (V)	1200	2200	2600	2750	2850

OCC for above generator

$\frac{2000}{6}$



Q2 A fully compensated 250-V 110-A dc Shunt motor has following parameters
 $R_A = 0.15 \Omega$ $N_F = 1500$ turns $N_A = 80$ turns
 $R_F = 200 \Omega$ $n_m = 1250$ r/min
 $R_{adj} = 10$ to 300Ω

Find out following

(a) Under no-load conditions, what range of possible speeds can be achieved by adjusting R_{adj}

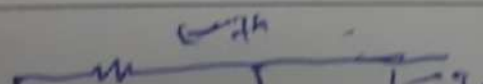
(b) What is full-load speed of this motor when $R_{adj} = 100 \Omega$?

(c) What is induced torque in part (b)

This motor has compensating windings and interpoles. The magnetization curve for this motor is shown in Figure below

Magnetization Curve for above motor at speed = 1250 rpm					
I_f (A)	0.25	0.5	0.75	1	1.25
E_A (V)	100	200	270	285	290

(a)



⊗

$E_A = V_T$ at no load

- Q3 Objective type: short answers, diagrams etc [Binary Marking]
- 1) Draw the torque speed curve of a compensated and uncompensated DC shunt motor on the same coordinate axis [20] CLO
 - 2) A 250 V 150A DC shunt motor has $R_A = 0.1 \text{ ohm}$ and $R_F = 250 \text{ ohm}$, it has a three stage starter circuit consisting of three equal resistances, what should be the value of each resistance if the starting current at the time of switch closing should not exceed double the full load current. 03 04
 - 3) Draw the torque speed curve of a DC Series Motor
 - 4) Draw the phasor diagram of an alternator to show the effect of change of inductive load keeping I_F constant (ignore R_A) (begin)
 - 5) A delta connected synchronous generator was tested for a DC test. When a DC voltage of 10 V was applied to two of the terminals, a current of 30 A was measured. What is the value of per phase armature resistance.

1) torque

